

EDUGENIE

Google Gemini Powered Learning Assistant

5. Project Development

Backend Development

- The backend is implemented with FastAPI.
- Each major learning feature is organized into a dedicated Python module.
- RESTful endpoints connect frontend requests to the appropriate module logic.

Explanation Module

- The documented design uses LaMini-Flan-T5-783M for concise, context-aware explanations.
- The objective is to break down complex topics into language that is easier for beginners and self-learners to understand.

Q&A; Module

- The Q&A; functionality uses Gemini for general knowledge and academic question-answering.
- The module is designed to provide contextual AI assistance across different educational topics.

Quiz Module

- The module requests three multiple-choice questions from a supplied passage.
- Each question contains four options and a correct answer.
- The documented implementation expects structured JSON and cleans Markdown code blocks before parsing.

Summary and Learning Path Modules

- The summary module produces concise versions of longer educational passages while retaining key information.
- The learning-path module asks Gemini to organize beginner-to-advanced concepts and suggest resources such as videos, articles, or books.

Frontend Integration

- The HTML form provides task selection and text input.
- Form submission sends a POST request to the FastAPI backend.
- The generated result is displayed below the input area.

Source basis: EduGenie project documentation provided by the user.